Chapter 6 Critical Thinking

Textbook page 143: questions 2–9 and 11

2. Method declaration and body

The declaration identifies the method, its access level, return type, name, and parameters. The body is the code inside its braces that carries out the task.

3. Access level

An access modifier, such as public or private, sets a method's access level.

4. Another term

The access level is also called visibility.

5. Variable scope

var1 is local to main, from its declaration to the end of main.

var2 is local to the for loop in main, including its condition, update, and body.

var3 is local to method1, from its declaration to the end of method1.

var4 is local to the for loop in method1.

As printed, the example also has errors: var3 is used before initialization, main is outside the class because of an early closing brace, and method() does not match method1().

6. Method declarations

a) public static int getVowels(String text)

b) public static int extractDigit(int number)

c) public static String insertString(String text, int position)

7. Distinguishing methods

a) The compiler uses the method signature: its name and the number, types, and order of its parameters.

b) Yes. This is overloading. The parameter lists must differ; changing only the return type is not enough.

8. Return statements

a) return ends the method and can send its result back to the caller.

b) A return statement sends back one value.

c) A method returning a value declares its return type, such as int. A method returning no value declares void.

9. Error in MethodCallExample

The class is closed before main, so main is outside the class. Move that closing brace to after main. The first doSomething() call is legal but discards the returned 5. The second stores 5 in num.

11. True or false

a) True.

b) False. A method call uses the method name followed by arguments in parentheses. It does not repeat the declaration.

c) False. A void method does not return a value.

d) False. An access modifier controls visibility. The return type says what kind of value comes back.

e) True.

f) False. Parameters are enclosed in parentheses.

g) False. A local variable can only be used inside the block where it is declared, after its declaration.

h) False for primitive arguments. Assigning to the parameter changes the method's local copy, not the caller's variable. An object reference is also copied; methods can change the referenced object's contents.

i) False. Overloading means using the same method name with different parameter lists.

j) True.

k) False. Preconditions describe assumptions the caller must satisfy. The declaration already states parameter types.

l) False. A postcondition describes the result after the method finishes, rather than how it produces that result.